



Chemical Bonding in Materials

Jacques K. Desmarais

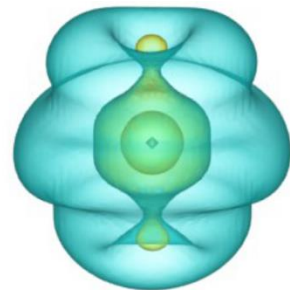
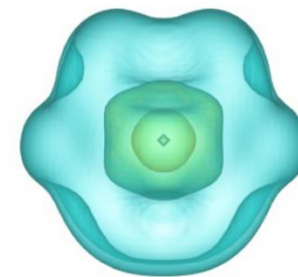
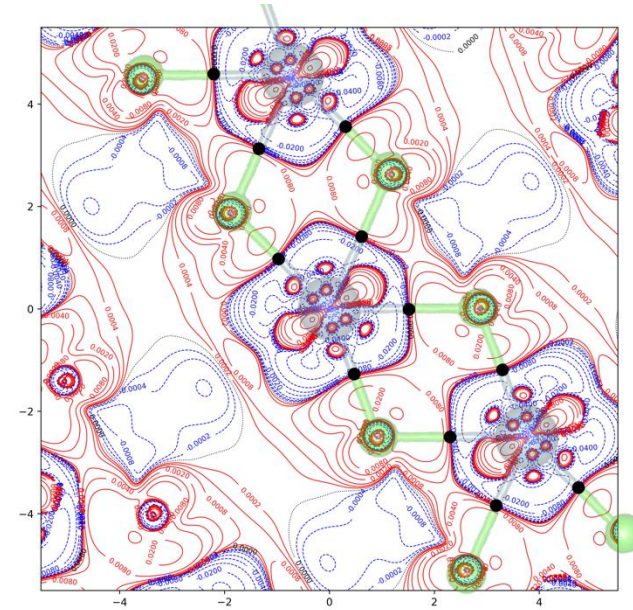
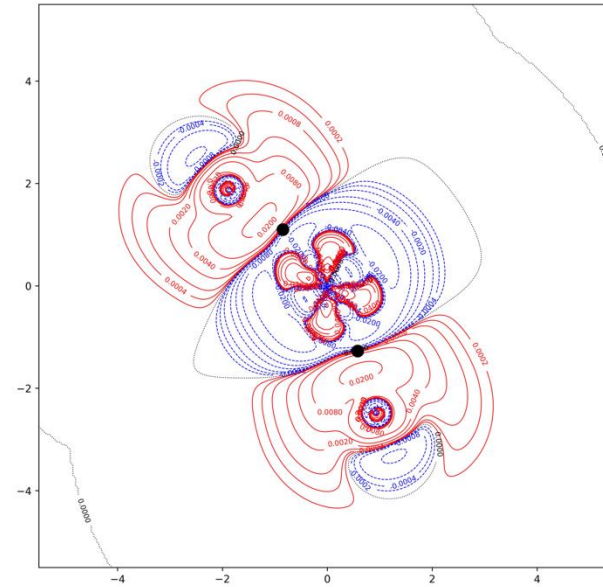
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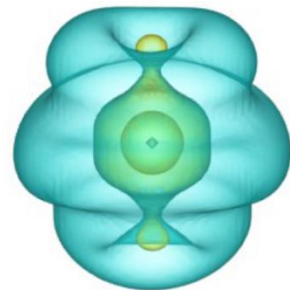
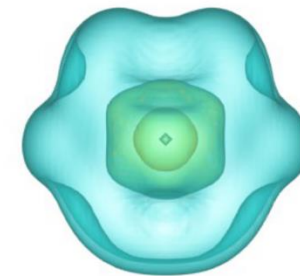
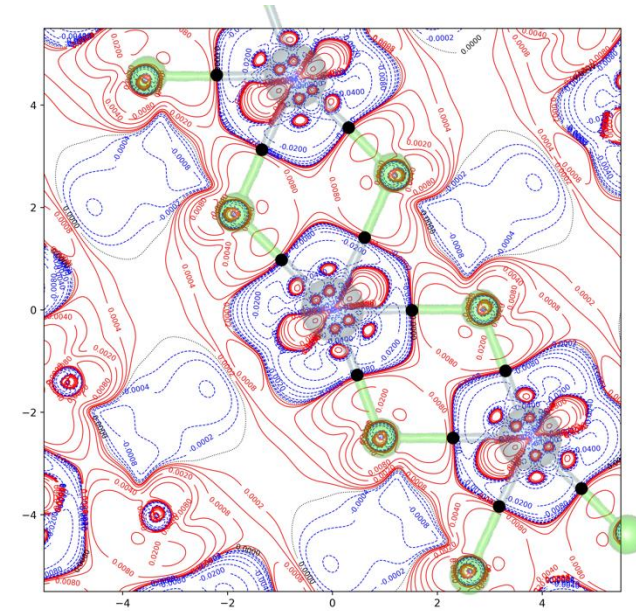
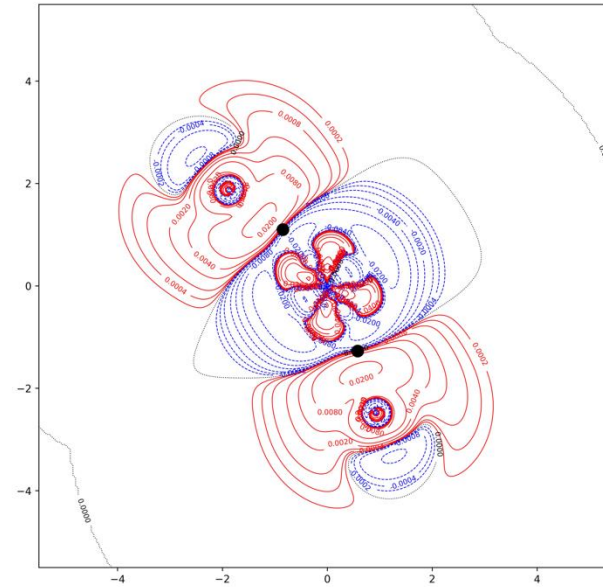
Outline

- Why?



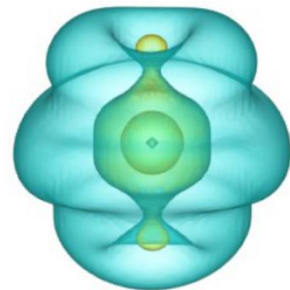
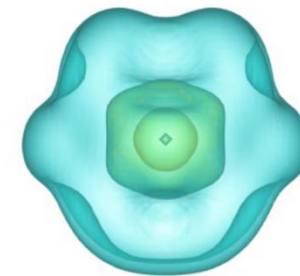
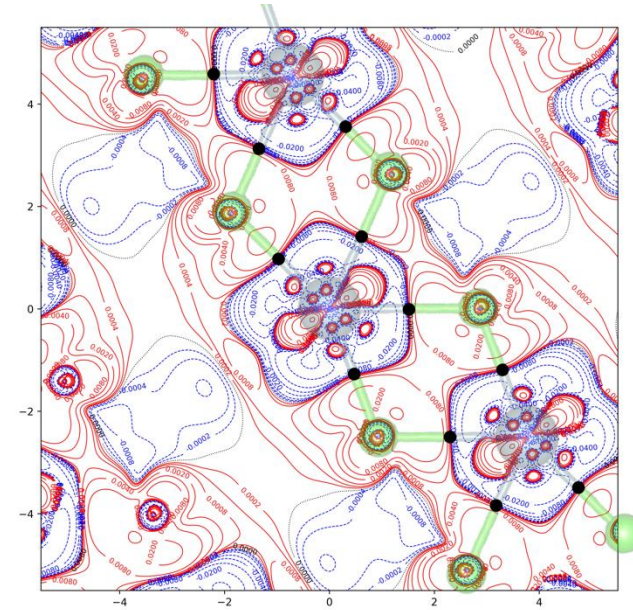
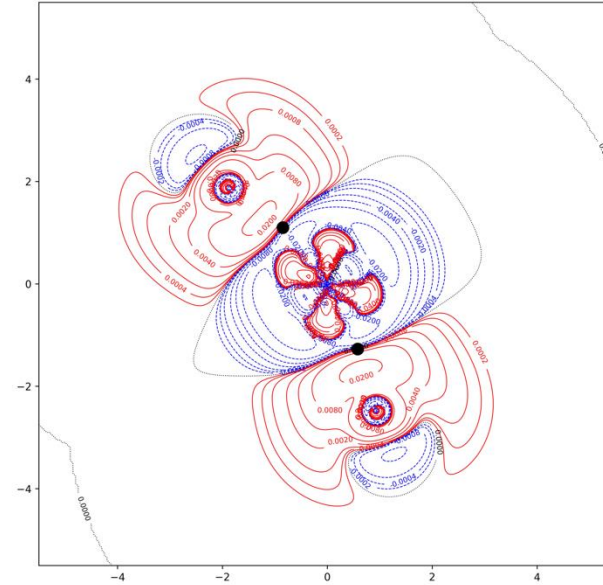
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- Why?
- The electron density in quantum mechanics



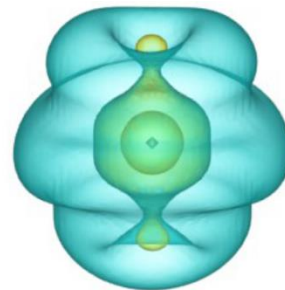
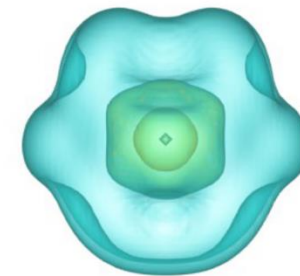
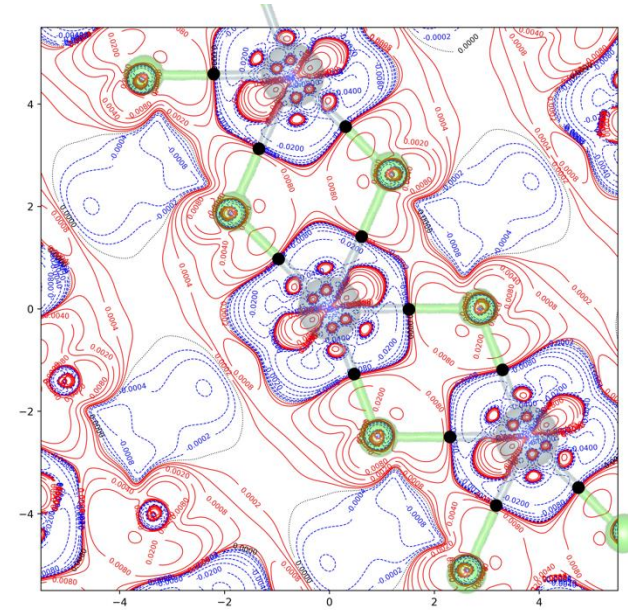
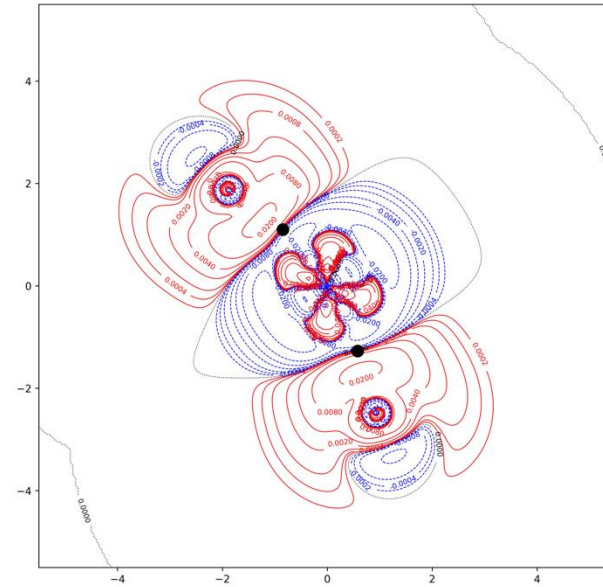
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- The electron density in quantum mechanics
- Topological analysis of the electron density



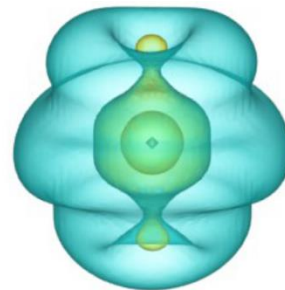
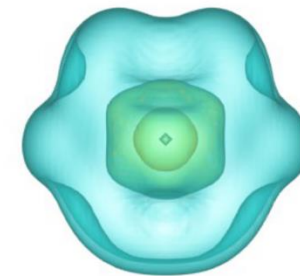
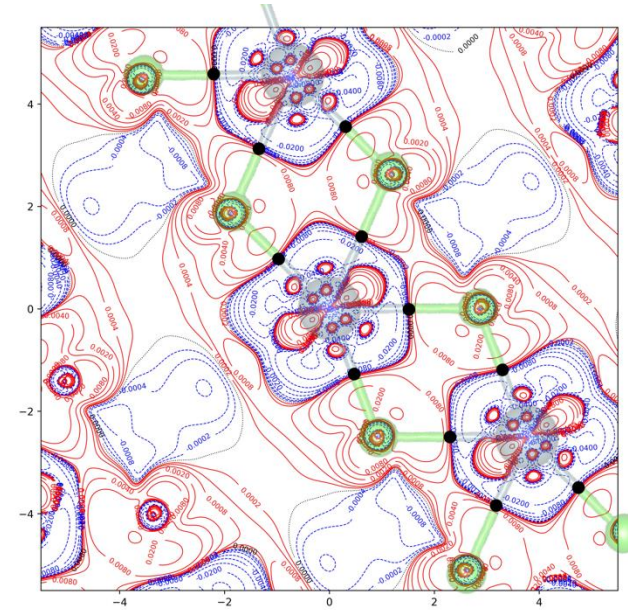
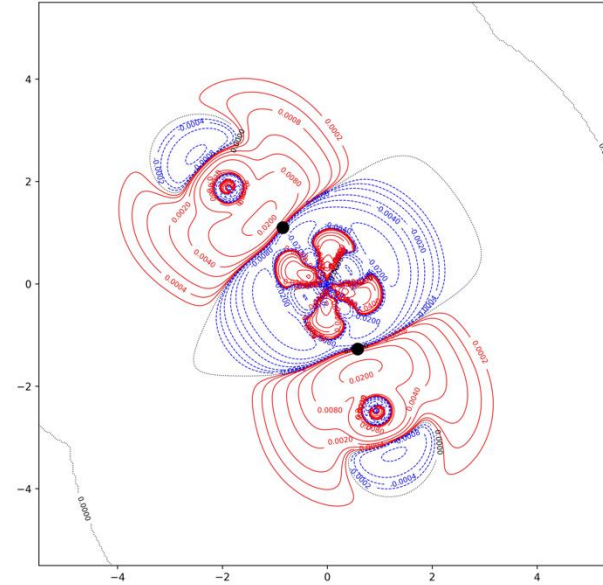
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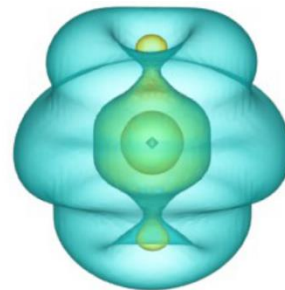
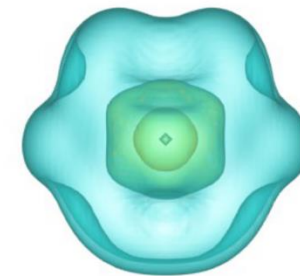
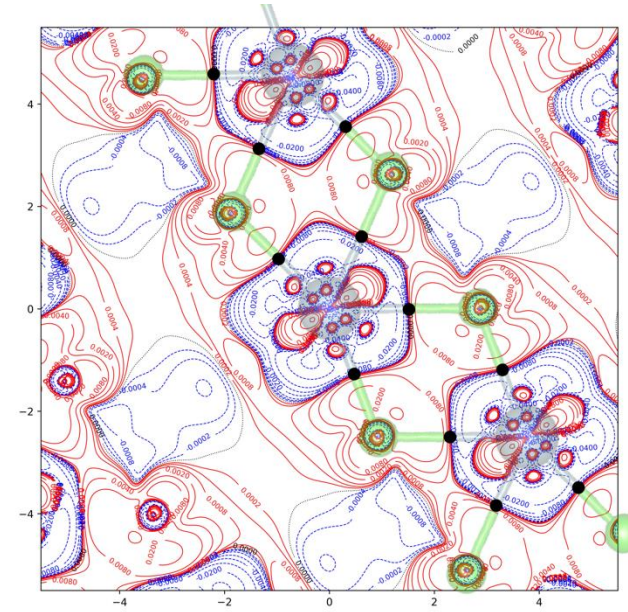
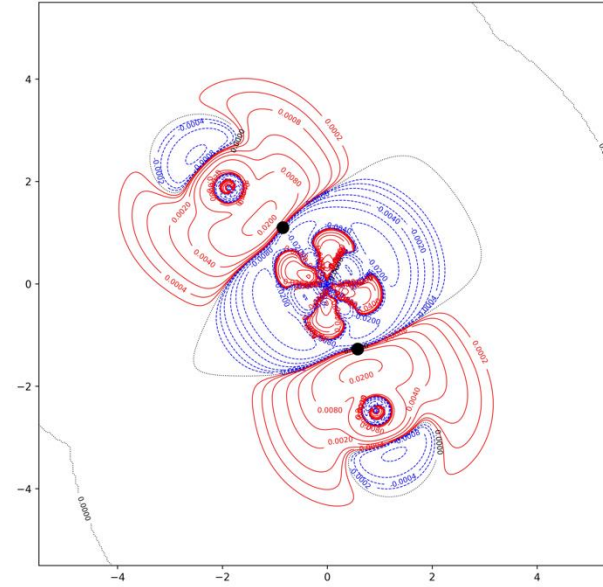
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- The Electron Localization Function



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- But experimentalists speak in a weird language that includes the terms:
 - « bonding »
 - « periodic table »
 - « oxidation state »
 - « acid », « base »
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- which intrinsically requires the concept « atom »
- How very odd and annoying!



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 2. I need to speak the same language as them: *I need a quantum theory of atoms!*



Electron Density in QM

- We are concerned with solving and analyzing the solutions of:

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- In the position representation:

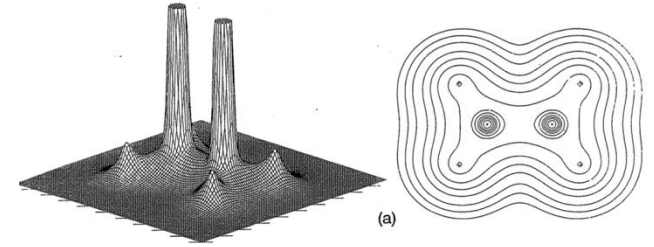
$$\begin{aligned} \Psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N) &= \langle \mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N | \vartheta_1, \vartheta_2, \dots, \vartheta_N \rangle \\ &= \frac{1}{\sqrt{N!}} \sum_{\varsigma \in \mathcal{S}_N} \{\text{sgn}(\varsigma) \vartheta_{\varsigma_1}(\mathbf{r}_1) \times \vartheta_{\varsigma_2}(\mathbf{r}_2) \times \dots \times \vartheta_{\varsigma_N}(\mathbf{r}_N)\} \end{aligned}$$

- It depends on the coordinates of all of the electrons



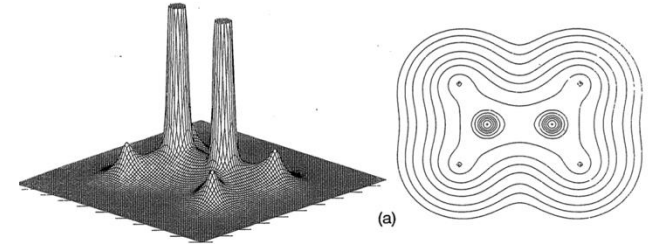
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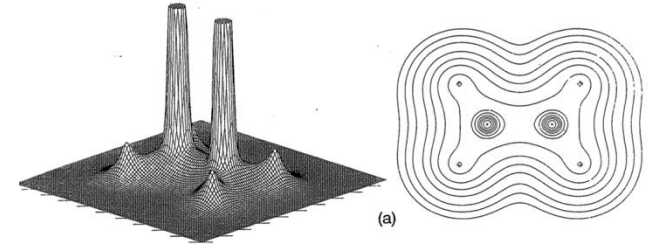


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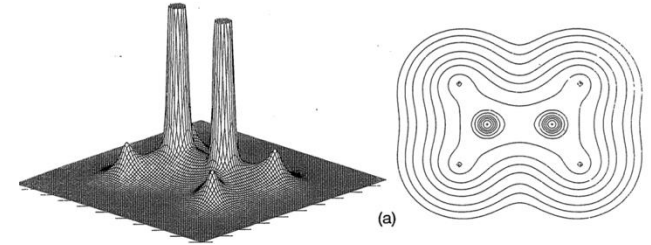
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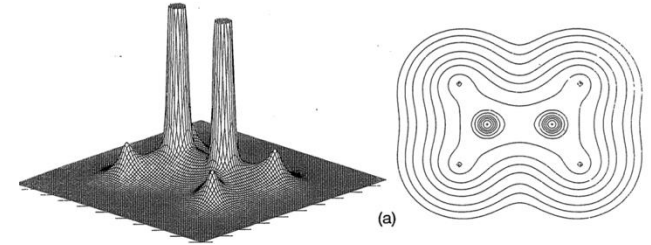
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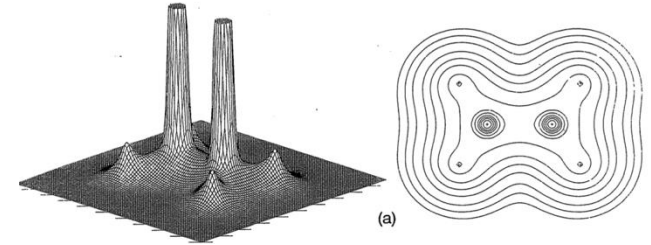


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$$\rho(\mathbf{r}) = N \int \cdots \int |\Psi(\mathbf{r}, \mathbf{r}_2, \dots, \mathbf{r}_N)|^2 d\mathbf{r}_2 d\mathbf{r}_3 \dots d\mathbf{r}_N$$



Electron Density in QM

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Inhomogeneous Electron Gas*

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University of California at San Diego, La Jolla, California

(Received 18 June 1964)

This paper deals with the ground state of an interacting electron gas in an external potential $v(\mathbf{r})$. It is proved that there exists a universal functional of the density, $F[n(\mathbf{r})]$, independent of $v(\mathbf{r})$, such that the expression $E \equiv \int v(\mathbf{r})n(\mathbf{r})d\mathbf{r} + F[n(\mathbf{r})]$ has as its minimum value the correct ground-state energy associated with $v(\mathbf{r})$. The functional $F[n(\mathbf{r})]$ is then discussed for two situations: (1) $n(\mathbf{r}) = n_0 + \tilde{n}(\mathbf{r})$, $\tilde{n}/n_0 \ll 1$, and (2) $n(\mathbf{r}) = \varphi(\mathbf{r}/r_0)$ with φ arbitrary and $r_0 \rightarrow \infty$. In both cases F can be expressed entirely in terms of the correlation energy and linear and higher order electronic polarizabilities of a uniform electron gas. This approach also sheds some light on generalized Thomas-Fermi methods and their limitations. Some new extensions of these methods are presented.



Topological analysis of the electron density

- For electrons in a Coulomb field, any observable can be written:

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- Richard Bader:
 - Atoms can be uniquely defined based on ρ
 - Provides the bridge between quantum mechanics and concepts of chemistry
 - A theoretical framework where « quantum mechanics » meets « bonding », « acid », « base », « oxidation state », etc...

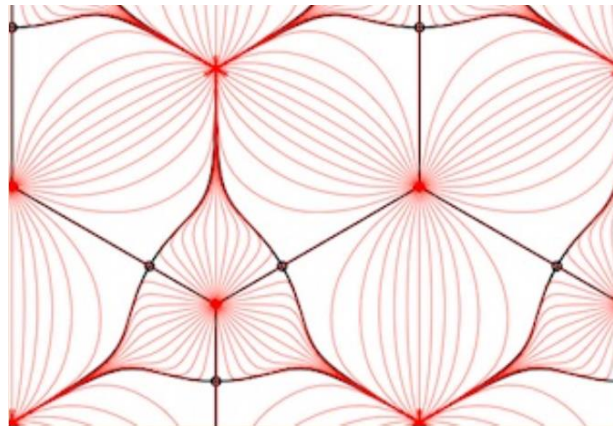
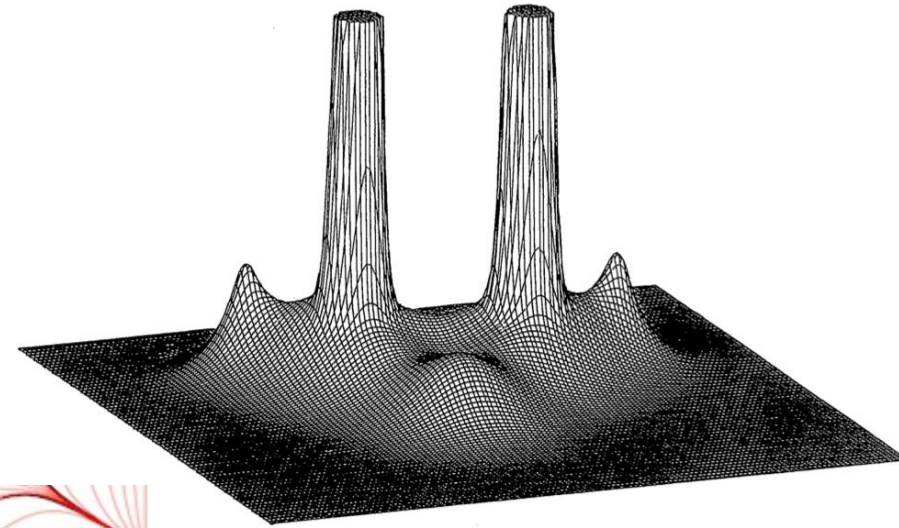


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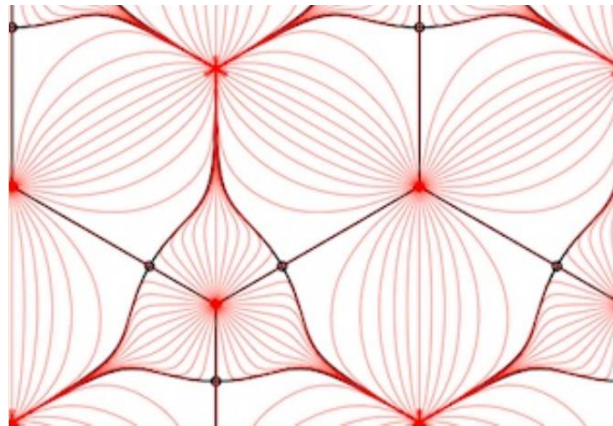
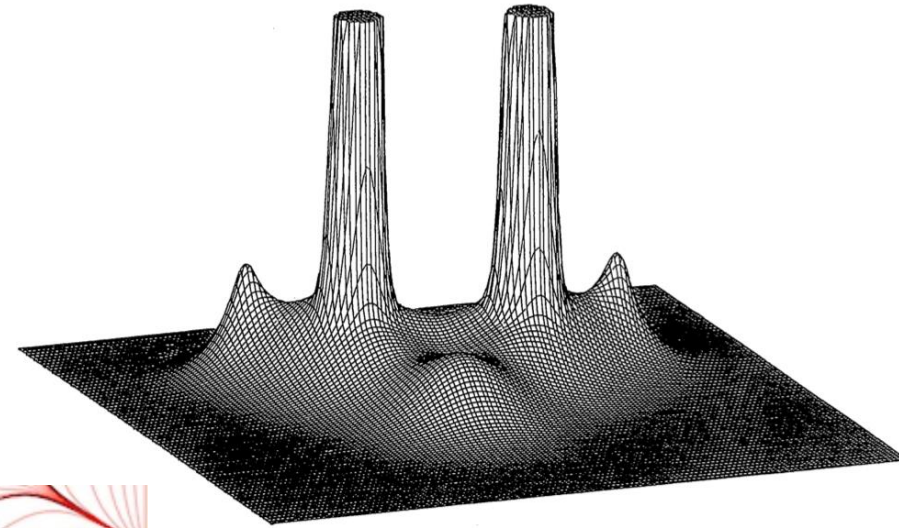
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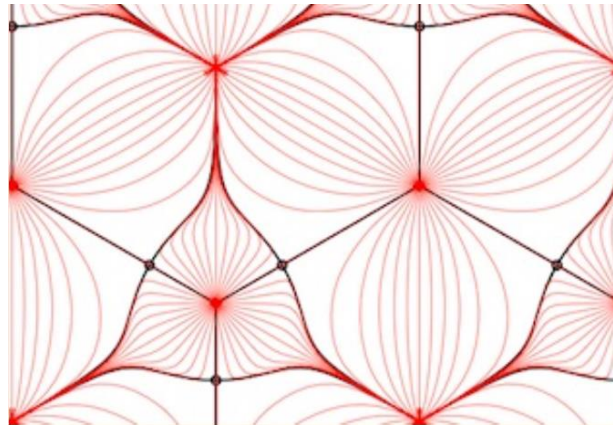
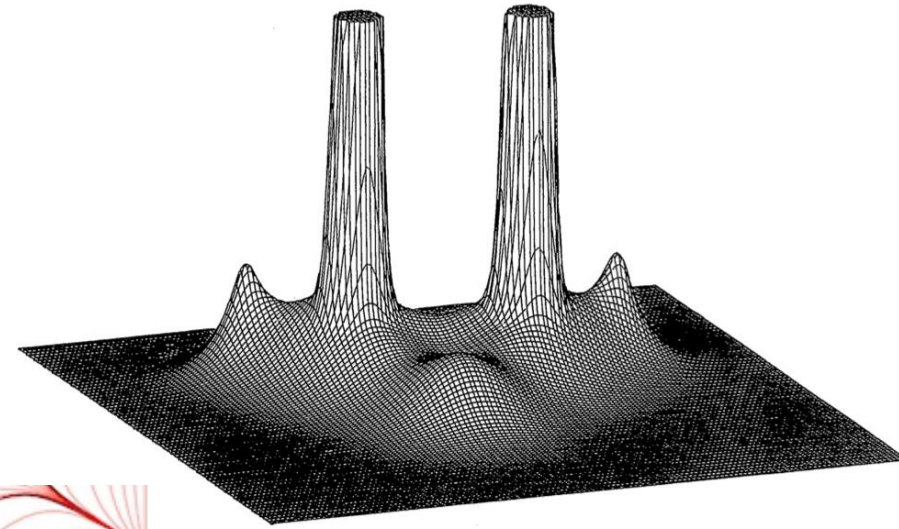
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- CPs: points $\mathbf{r}_c$ where $\nabla\rho(\mathbf{r}_c) = \mathbf{0}$
- Characterize the topology of ρ



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- Characterize the topology of ρ
- $\nabla\rho$ is a vector field
- Plotted along trajectory lines
- TLs are perpendicular to iso-lines of ρ



Topological analysis of the electron density

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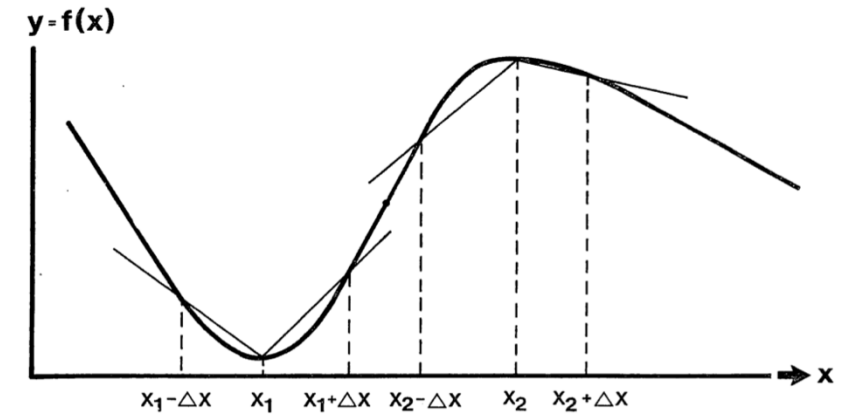


FIG. 2.2. Definition of curvature as the limiting difference ($\Delta x \rightarrow 0$) in the tangent lines which bracket a given point; at x_1 where $f(x)$ is a minimum and the curvature is positive and at x_2 where $f(x)$ is a maximum and the curvature is negative.



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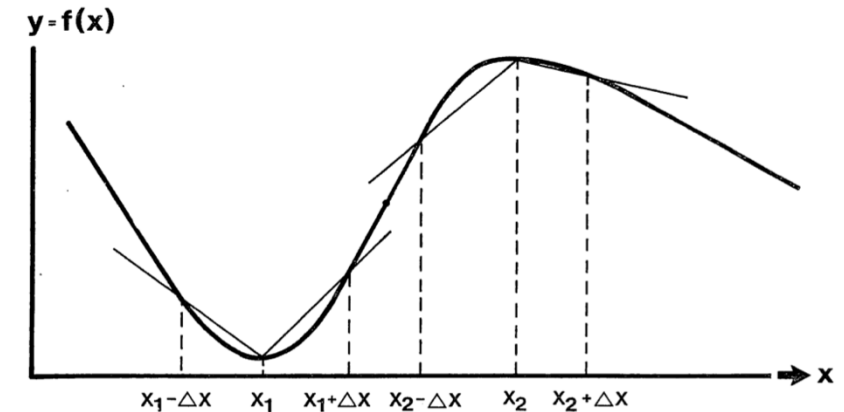


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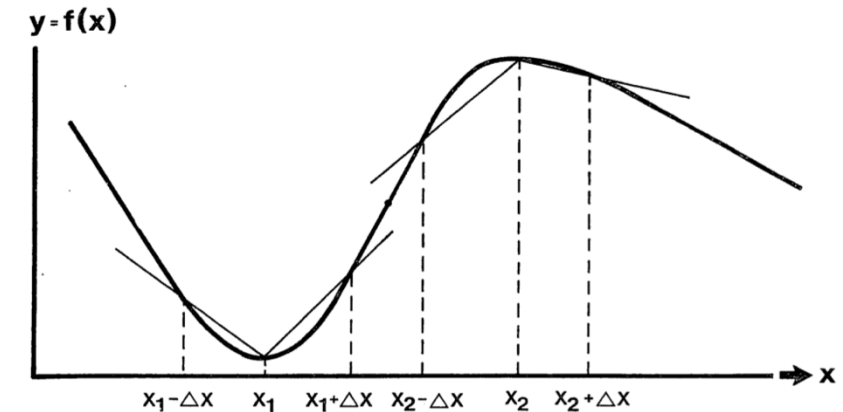


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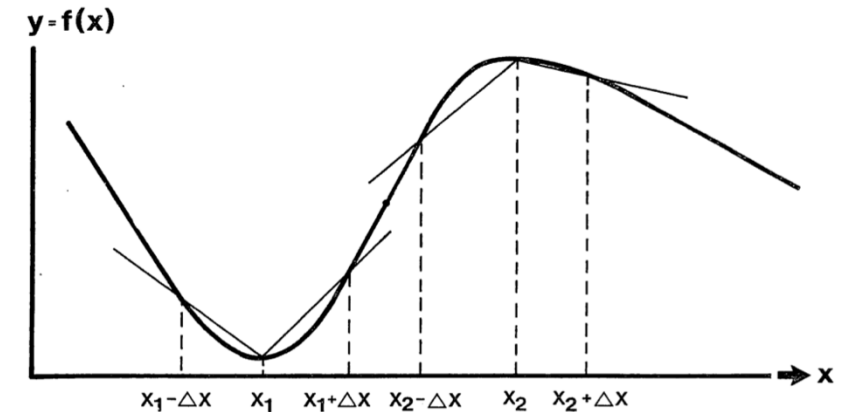


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Topological analysis of the electron density

- (RANK = # of non-zero λ_i , SIGNATURE = # of positive - # of negative λ_i)

• (3,-3) $0 > \lambda_1 \geq \lambda_2 \geq \lambda_3$ local maximum nuclei

• (3,-1) $\lambda_1 > 0 > \lambda_2 \geq \lambda_3$ Minimum along the axis joining two nuclei bond

• (3,+1) $\lambda_1 \geq \lambda_2 > 0 > \lambda_3$ Minimum along two directions ring

• (3,+3) $\lambda_1 \geq \lambda_2 \geq \lambda_3 > 0$ local minimum cage

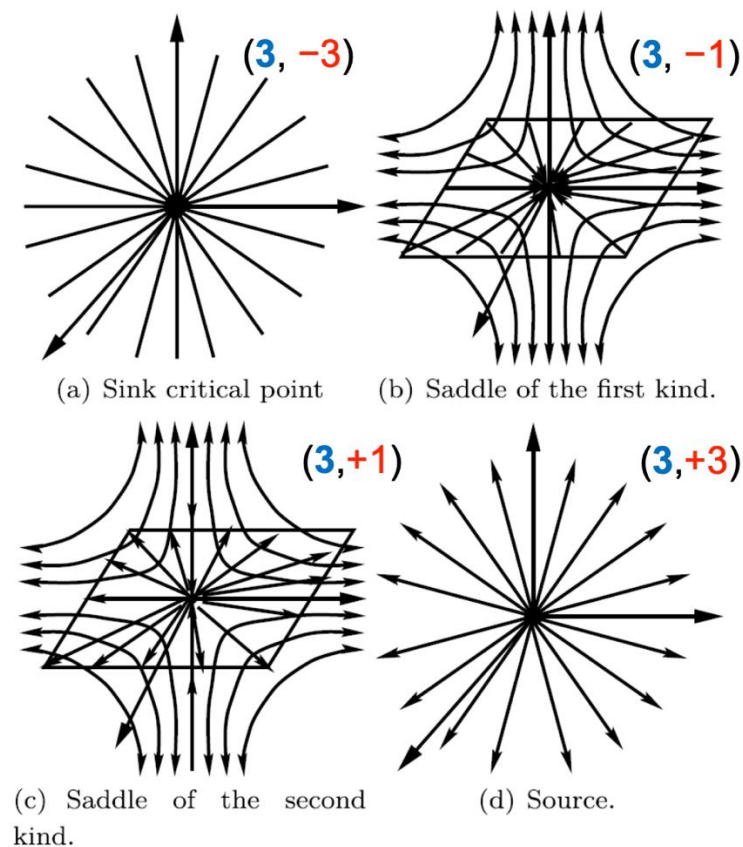
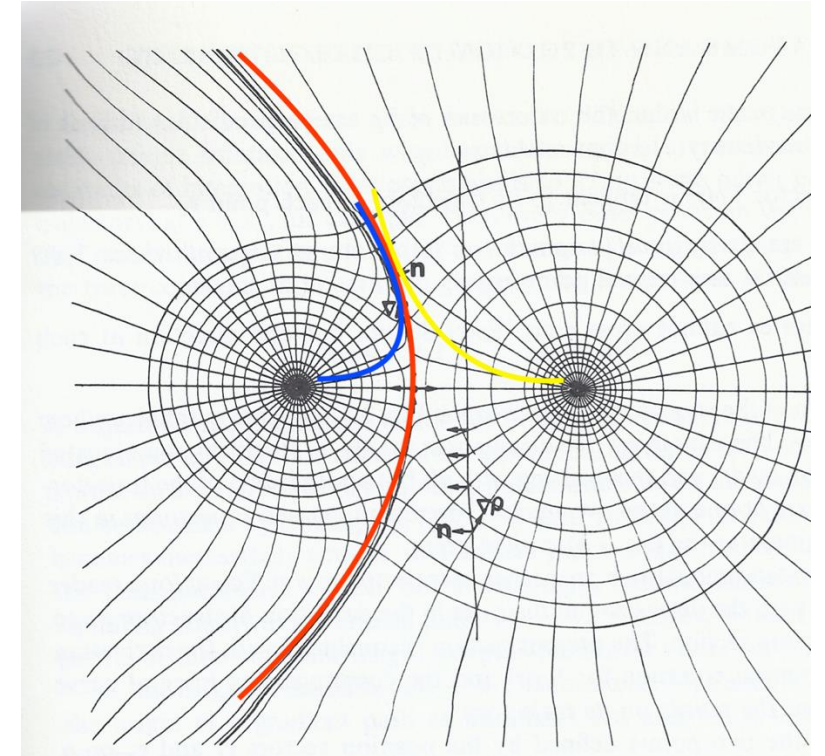


Figure 3.1: The four types of CPs of a 3D scalar field.
 Ángel Martín Pendás and Carlo Gatti, Ch. 3: Quantum theory of atoms in molecules and the AIMAll SW in *Complementary Bonding Analysis*, S. Grabowsky Ed., De Gruyter 2021



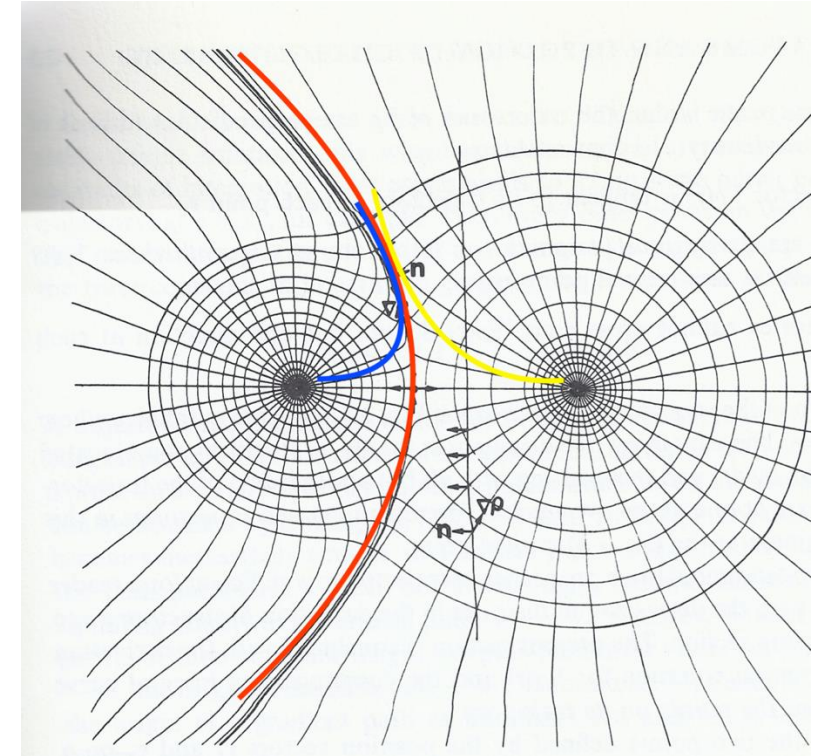
The quantum atom and bonding

- Bader's quantum atom:



The quantum atom and bonding

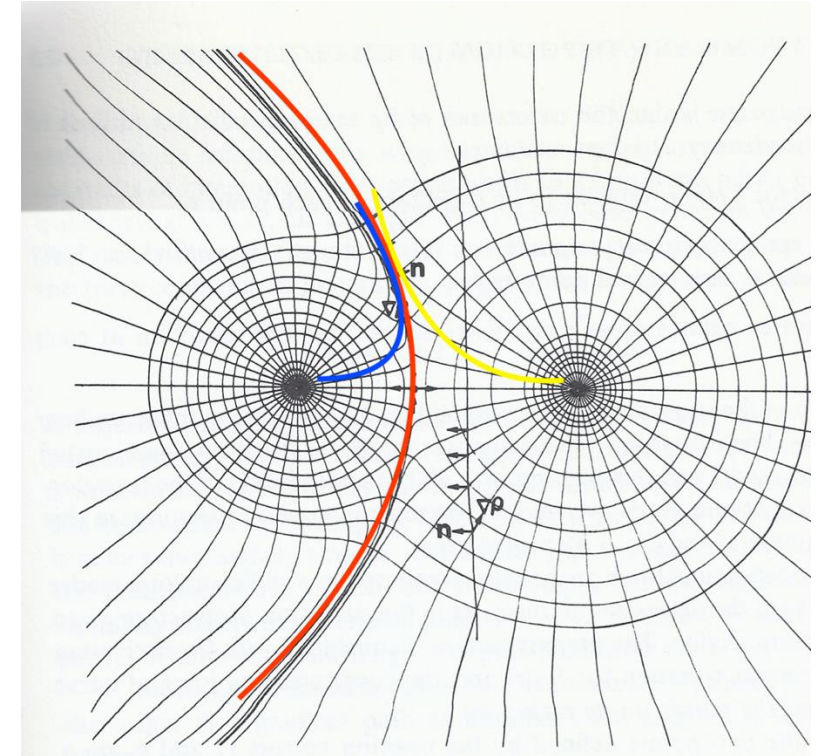
- Bader's quantum atom:
- Nuclei (i.e. $3,-3$ CP's) are charge attractors



The quantum atom and bonding

- Bader's quantum atom:
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- Basins are the spaces bordered by zero-flux surfaces:

$$\nabla \rho(\mathbf{r}) \cdot \mathbf{n}(\mathbf{r}) = 0 \quad \forall \mathbf{r} \in S(\Omega)$$

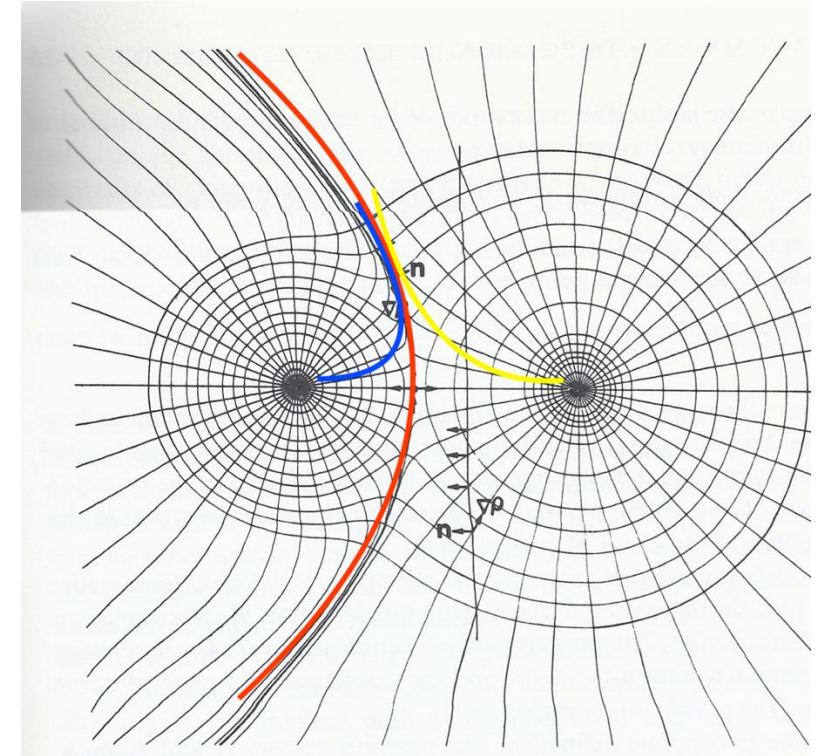


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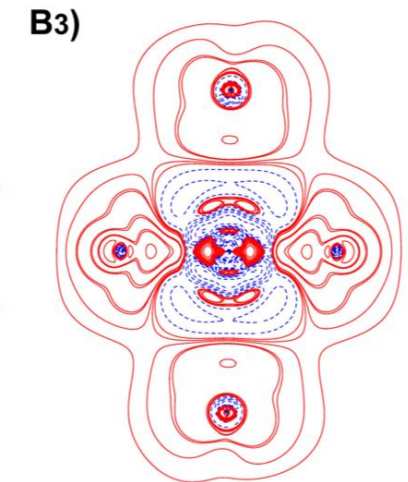
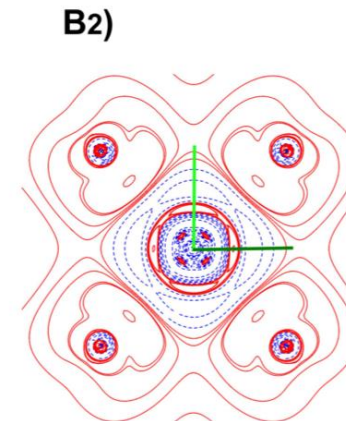
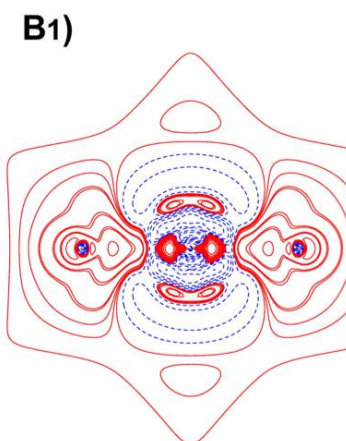
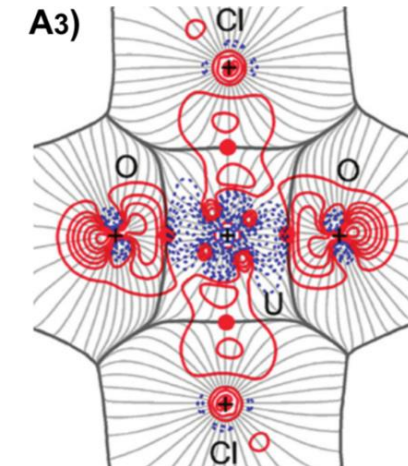
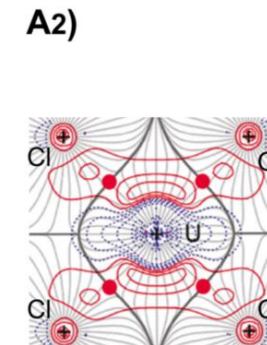
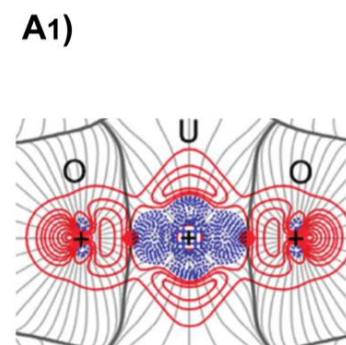
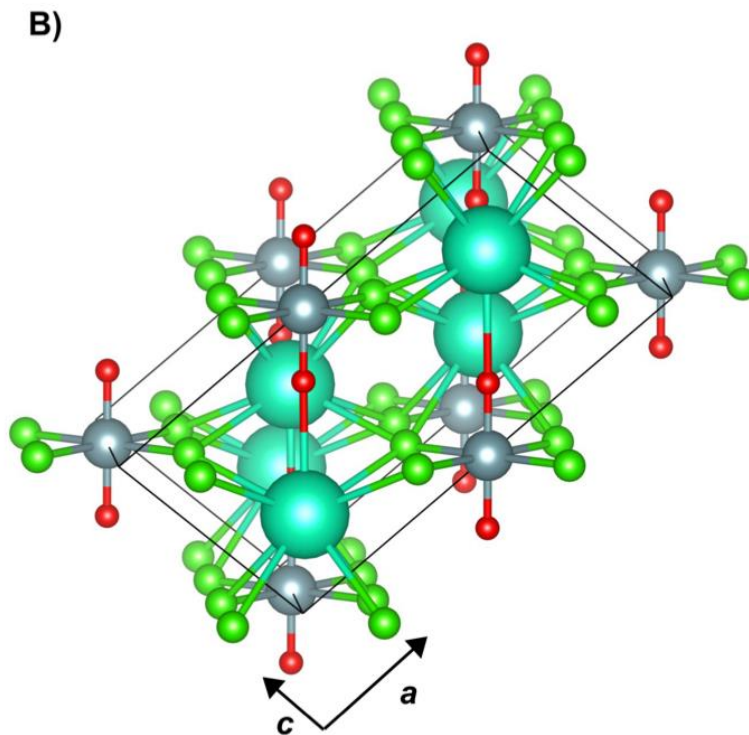
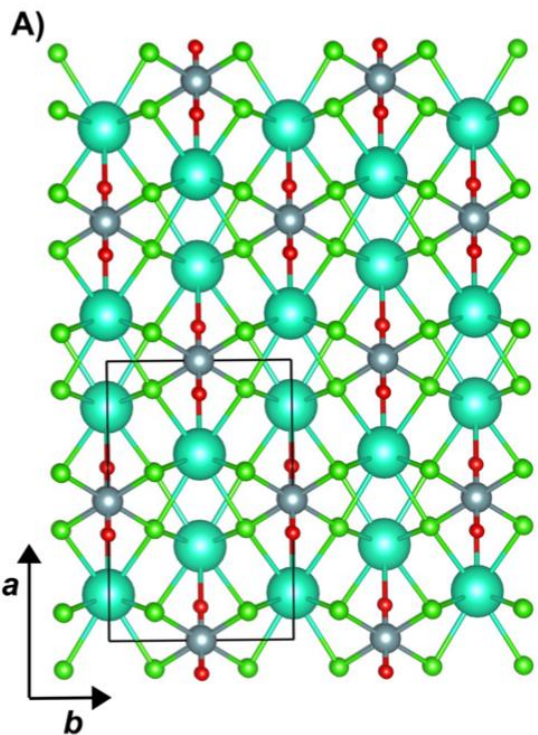
$$\nabla \rho(\mathbf{r}) \cdot \mathbf{n}(\mathbf{r}) = 0 \quad \forall \mathbf{r} \in S(\Omega)$$

- The union of an attractor with its basin is an atom!



The quantum atom and bonding

Topology of the density: ρ , $\nabla\rho$ and $\nabla\nabla\rho$



Cossard et al., 2021



The TOPOND program

- Calculation of ρ , $\nabla\rho$, $\nabla\nabla\rho$ (topological analysis of the electron density)
- Calculation of $\nabla \cdot \nabla\rho$, $\nabla(\nabla \cdot \nabla\rho)$, $\nabla\nabla(\nabla \cdot \nabla\rho)$ (topological analysis of the Laplacian)
- Search and Classification of CPs
- Calculation of local properties at the CPs and integrated over the atomic basin

TOPOND23

User's Manual

October 14, 2022

C. Gatti¹, S. Casassa², J.K. Desmarais², A. Cossard² and A. Erba²

- Now extended to work with f and g basis functions!
- E.g. with TOPOND23 it is now possible to study bonding in lanthanides and actinides



Electron localization function (ELF)

Classification of chemical bonds based on topological analysis of electron localization functions

B. Silvi & A. Savin

Laboratoire de Dynamique des Interactions Moléculaires, UPR271,
Université Pierre et Marie Curie, 4 Place Jussieu, 75252 Paris cédex,
France

THE definitions currently used to classify chemical bonds (in terms of bond order, covalency versus ionicity and so forth) are derived from approximate theories¹⁻³ and are often imprecise. Here we outline a first step towards a more rigorous means of classification based on topological analysis of local quantum-mechanical functions related to the Pauli exclusion principle. The local maxima of these functions define 'localization attractors', of which there are only three basic types: bonding, non-bonding and core. Bonding attractors lie between the core attractors (which themselves surround the atomic nuclei) and characterize the shared-electron interactions. The number of bond attractors is related to the bond multiplicity. The spatial organization of localization attractors provides a basis for a well-defined classification of bonds, allowing an absolute characterization of covalency versus ionicity to be obtained from observable properties such as electron densities.

NATURE · VOL 371 · 20 OCTOBER 1994

A simple measure of electron localization in atomic and molecular systems

A. D. Becke and K. E. Edgecombe

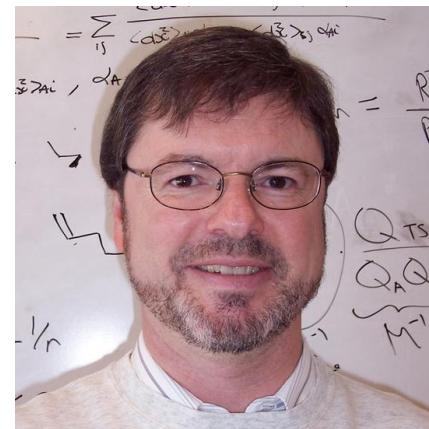
Department of Chemistry, Queen's University, Kingston, Ontario, Canada K7L 3N6

(Received 26 September 1989; accepted 24 January 1990)

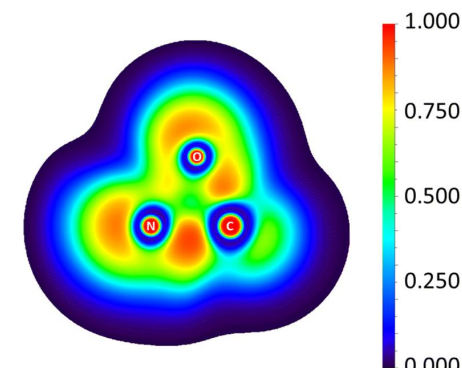
We introduce in this work a new approach to the identification of localized electronic groups in atomic and molecular systems. Our approach is based on local behavior of the Hartree–Fock parallel-spin pair probability and is completely independent of unitary orbital transformations. We derive a simple “electron localization function” (ELF) which easily reveals atomic shell structure and core, binding, and lone electron pairs in simple molecular systems as well.



Andreas Savin



Axel Becke



Michalski et al, 2019

Electron localization function (ELF)

$$\text{ELF}(\mathbf{r}) \equiv \frac{1}{1 + [D(\mathbf{r})/D^{\text{unif}}(\mathbf{r})]^2}$$

$$\tau = \frac{1}{2} \sum_j^{\text{occ}} |\nabla \varphi_j|^2$$

$$D(\mathbf{r}) = \tau(\mathbf{r}) - \frac{\nabla n(\mathbf{r}) \cdot \nabla n(\mathbf{r})}{8n(\mathbf{r})}$$

Measure different of fermionic kinetic energy w.r.t. boson kinetic energy

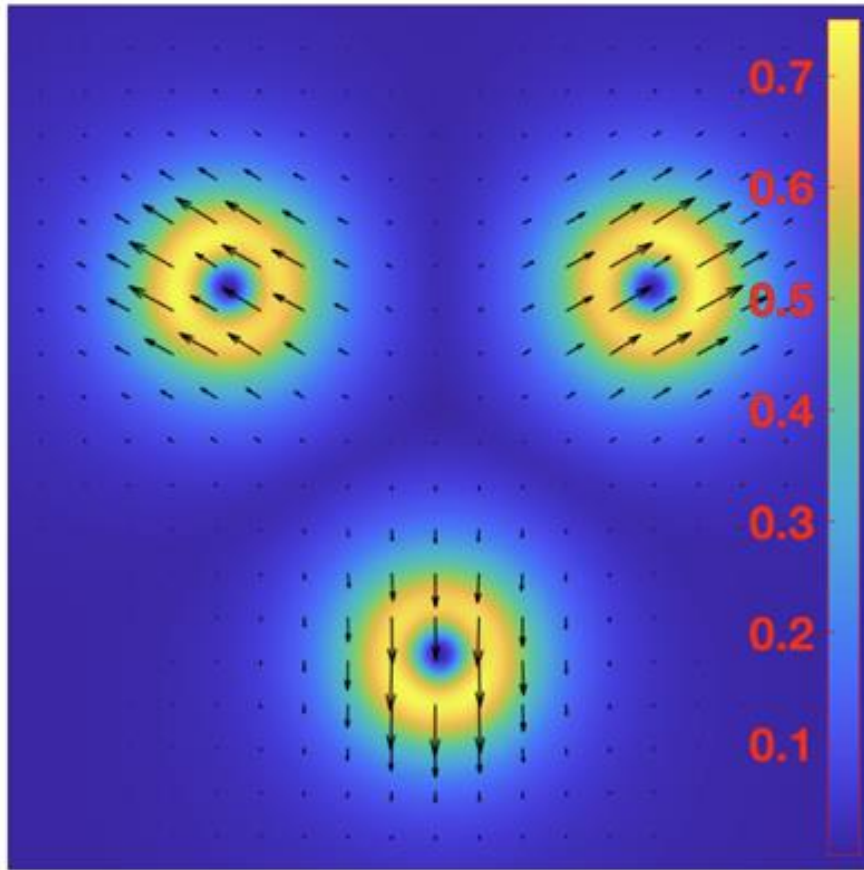
$$D^{\text{unif}}(\mathbf{r}) = \frac{3}{10} (3\pi^2)^{2/3} n^{5/3}(\mathbf{r})$$

Normalization relative to the uniform gas

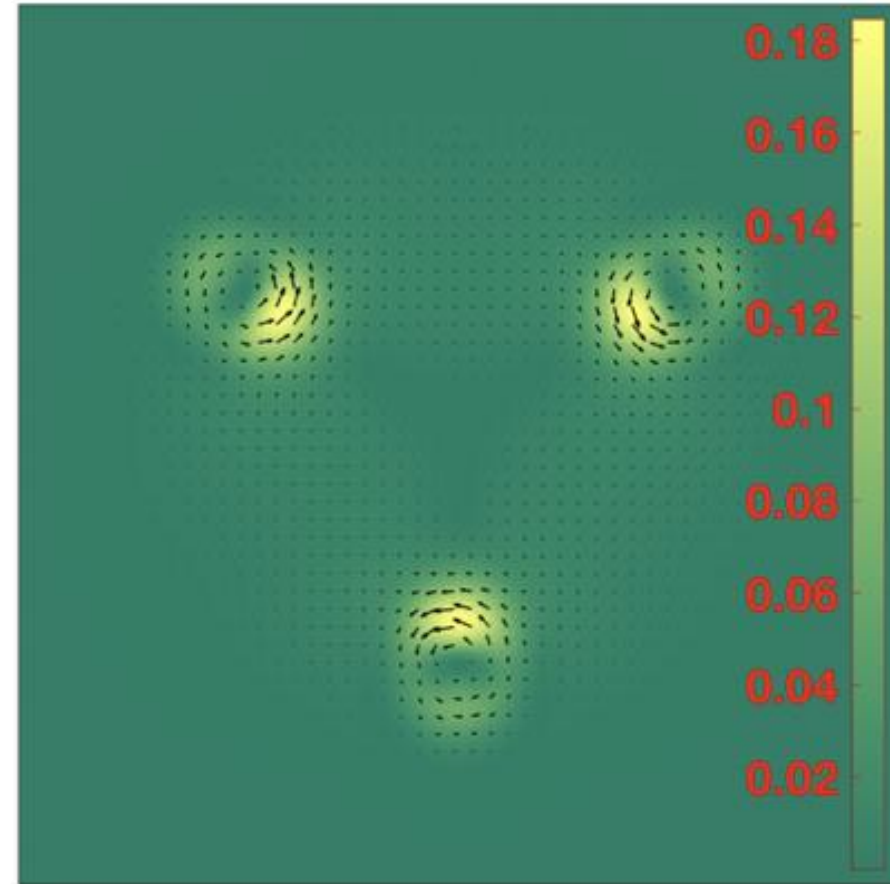
Slide courtesy of Dr. Stefano



What about magnetized systems?



magnetization

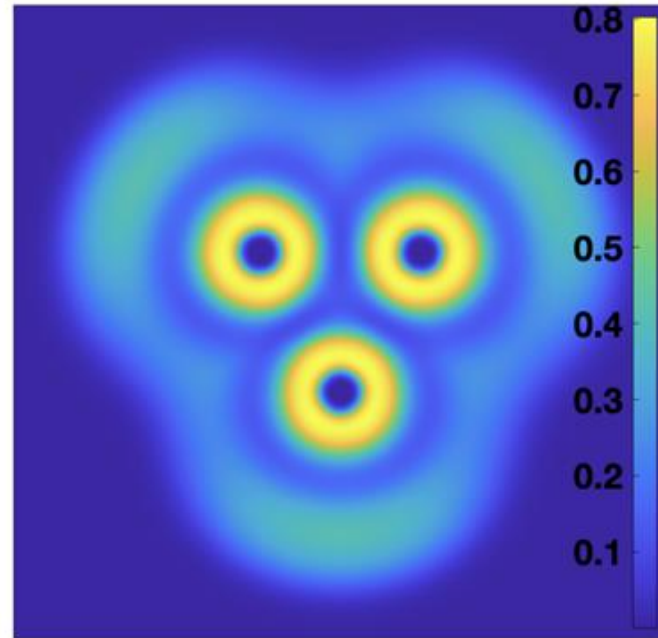


spin current

Slide courtesy of Dr. Stefano



Demonstration in Cr3



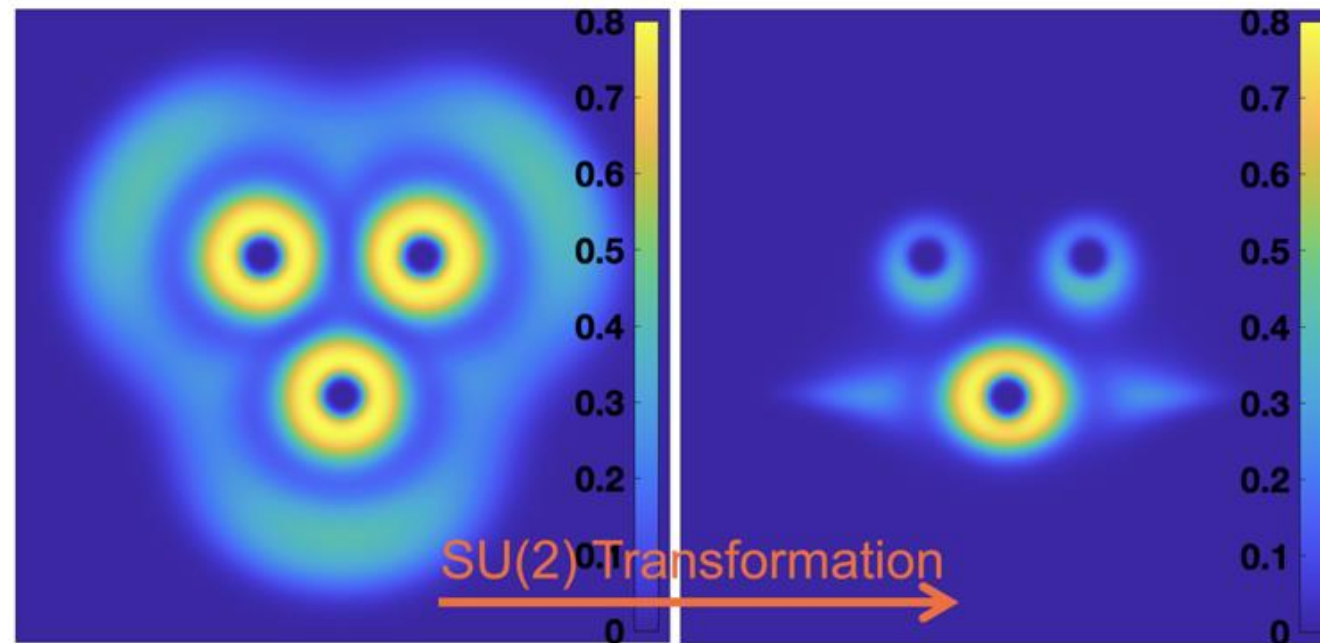
Slide courtesy of Dr. Stefano

x

J.K. Desmarais, MSSC2024, your computer screen



Demonstration in Cr3

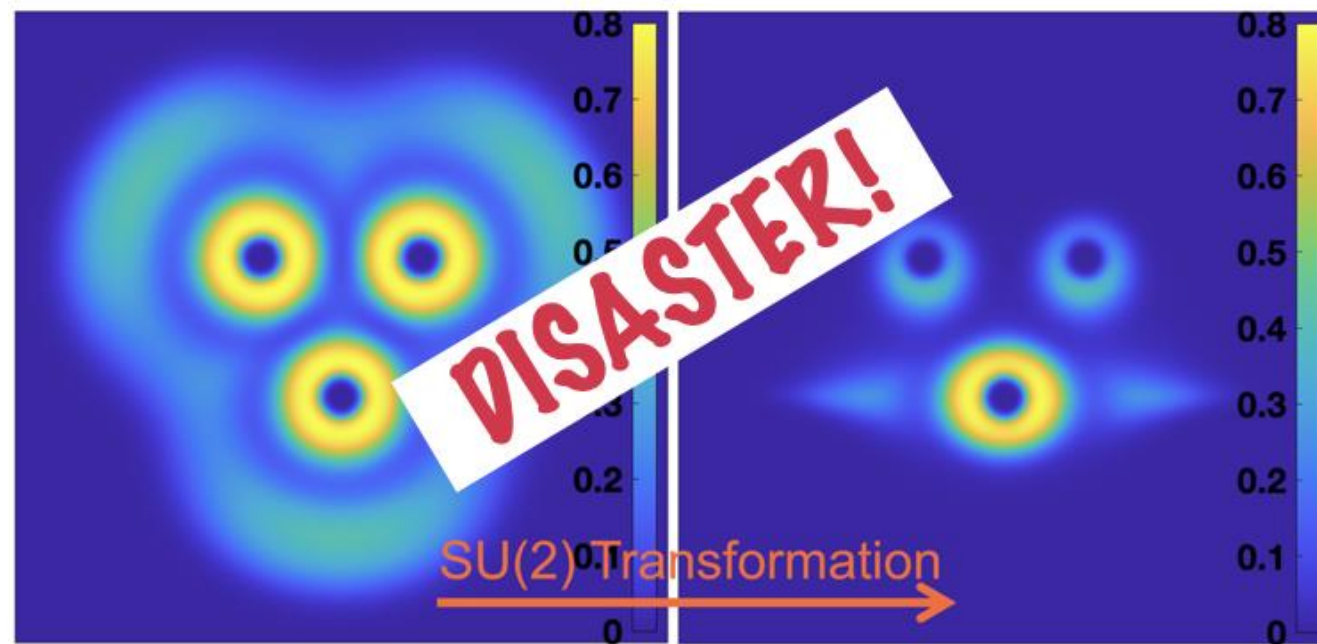


$$\begin{aligned} SU(2) : \quad \hat{\Psi}(\mathbf{r}) \rightarrow \hat{\Psi}'(\mathbf{r}) &= \exp \left[\frac{i}{c} \vec{\Lambda}(\mathbf{r}) \circ \vec{\sigma} \right] \hat{\Psi}(\mathbf{r}) \\ &= U_S(\mathbf{r}) \hat{\Psi}(\mathbf{r}) \end{aligned}$$

Slide courtesy of Dr. Stefano



Demonstration in Cr3



Physical quantities with direct meaning are gauge invariant!

Slide courtesy of Dr. Stefano



Electron Localization Function for Noncollinear Spins

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³*Setif 1 University-Ferhat Abbas, Faculty of Sciences, Department of Physics and Laboratory of Quantum Physics and Dynamical Systems, Setif, Algeria*

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(Received 24 May 2024; revised 14 July 2024; accepted 22 August 2024; published 25 September 2024)



Giovanni



Kamel



Alessandro



Stefano



$U(1) \times SU(2)$ Gauge Invariant ELF

$$\text{ELF}(\mathbf{r}) \equiv \frac{1}{1 + [D(\mathbf{r})/D^{\text{unif}}(\mathbf{r})]^2}$$

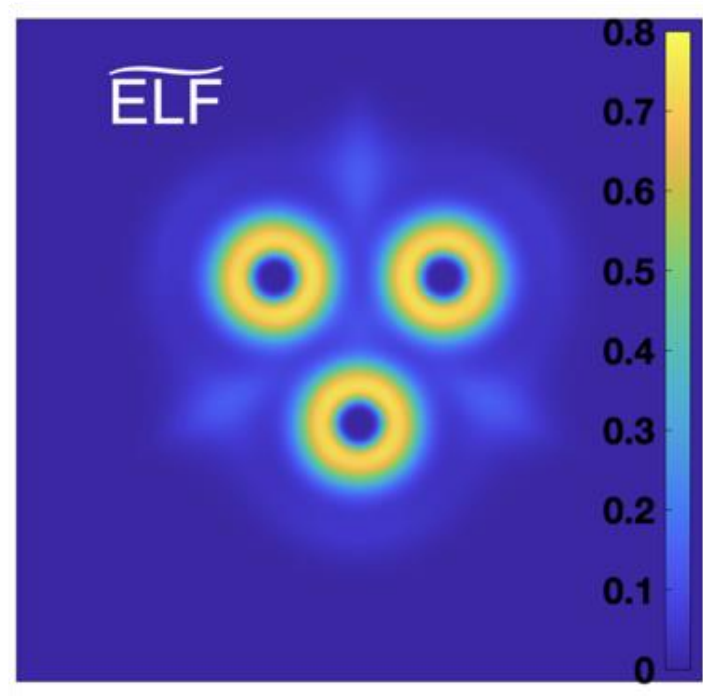

$$D(\mathbf{r}) = \tau(\mathbf{r}) - \frac{\nabla n(\mathbf{r}) \cdot \nabla n(\mathbf{r})}{8n(\mathbf{r})}$$


$$\tilde{\tau}_{\text{nc}}(\mathbf{r}) = \left(\tau_{\text{nc}}(\mathbf{r}) - \frac{\mathbf{j}(\mathbf{r}) \cdot \mathbf{j}(\mathbf{r})}{2n(\mathbf{r})} + \frac{\nabla m^a(\mathbf{r}) \cdot \nabla m^a(\mathbf{r})}{8n(\mathbf{r})} \right) + \left(\frac{m^a(\mathbf{r})\tau_{\text{nc}}^a(\mathbf{r})}{n(\mathbf{r})} - \frac{\mathbf{J}^a(\mathbf{r}) \cdot \mathbf{J}^a(\mathbf{r})}{2n(\mathbf{r})} \right)$$

Slide courtesy of Dr. Stefano



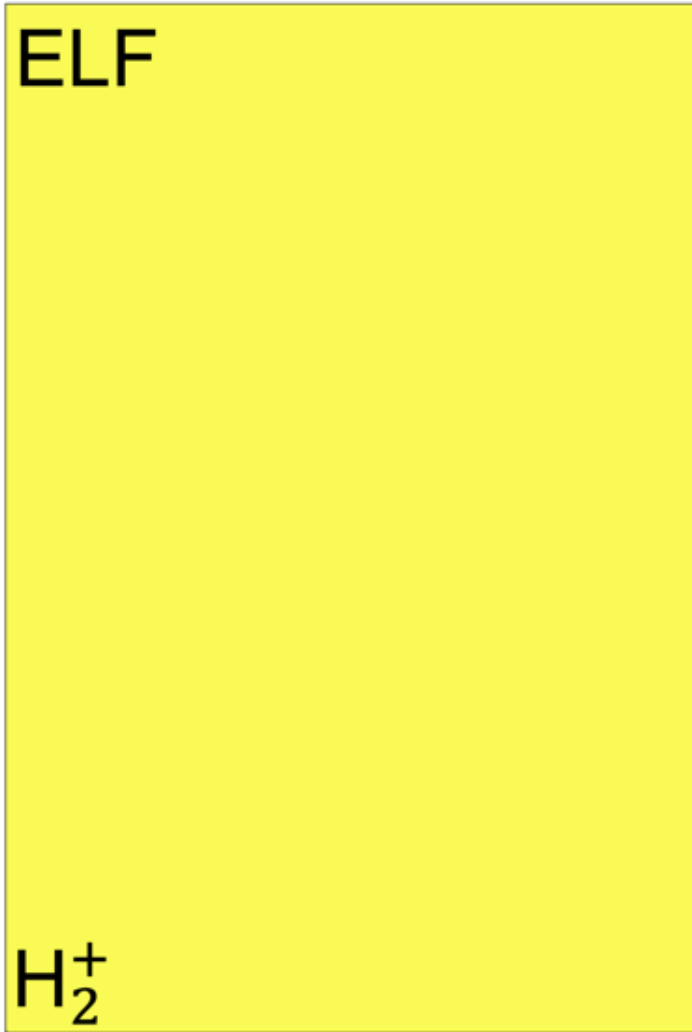
Demonstration in Cr3



Slide courtesy of Dr. Stefano



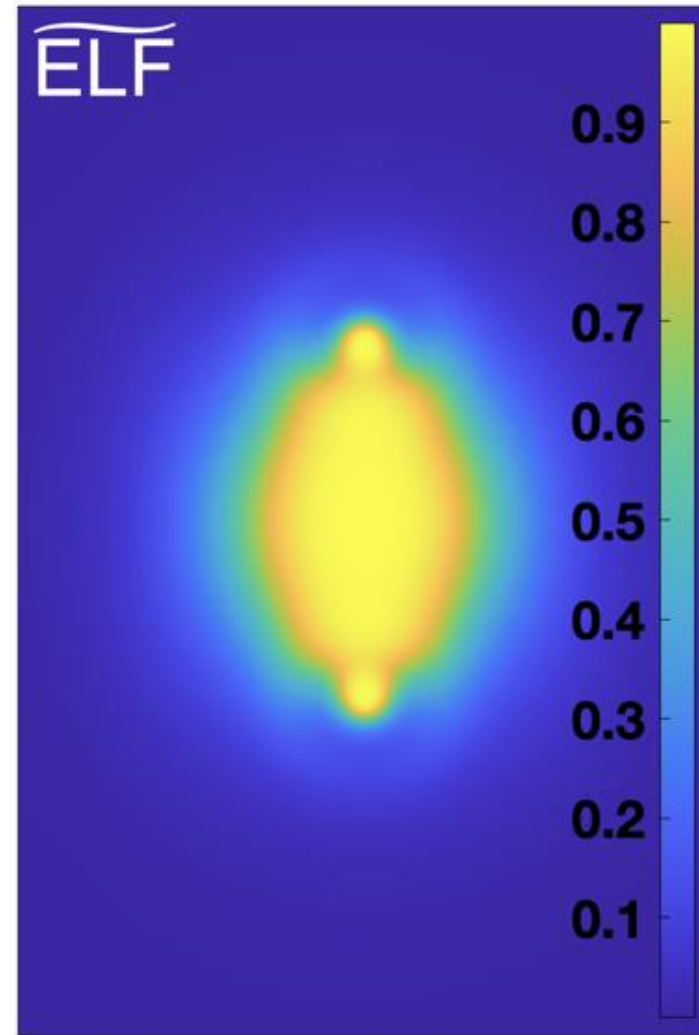
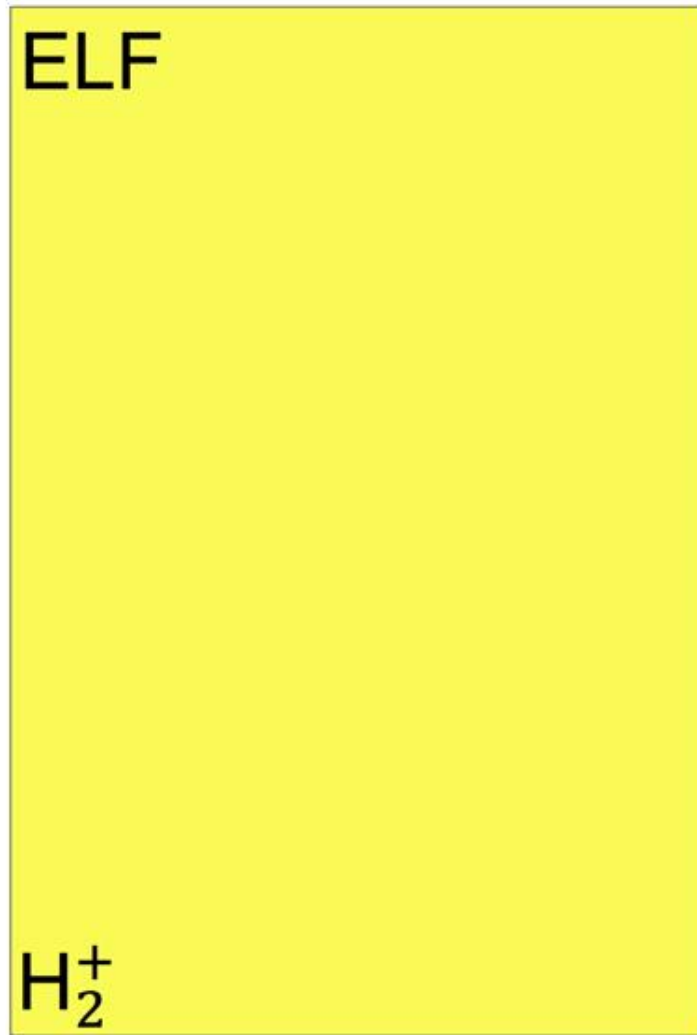
Bond in H_2^+



Slide courtesy of Dr. Stefano



Bond in H_2^+



Slide courtesy of Dr. Stefano



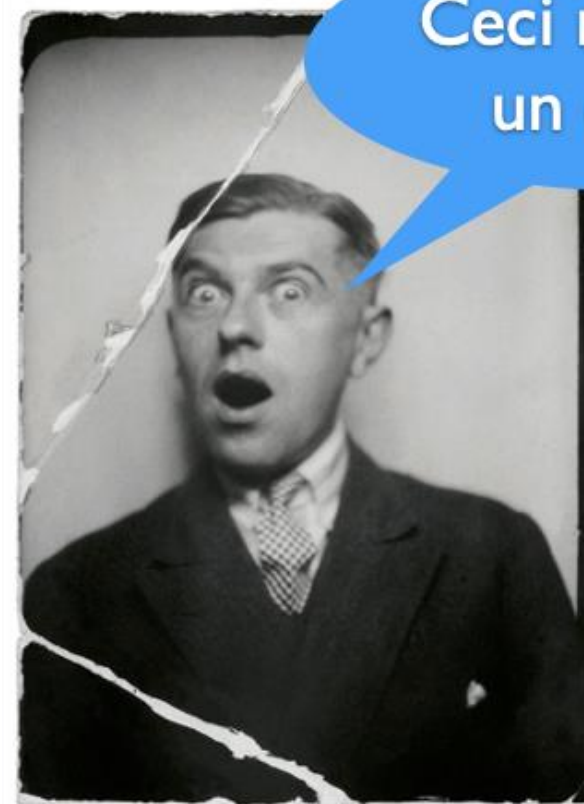
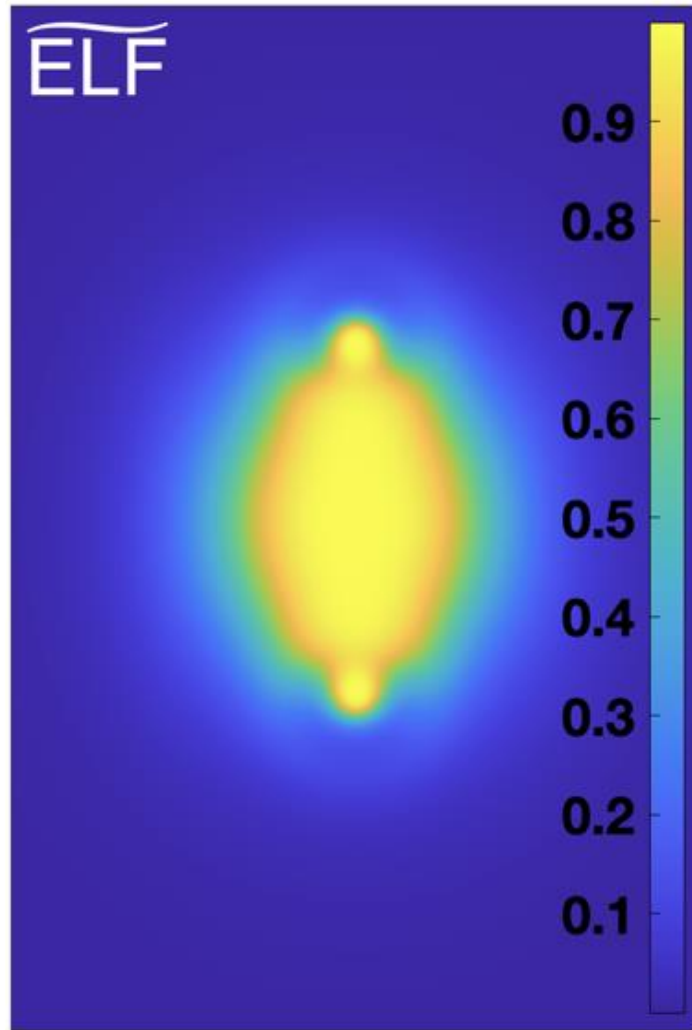
Bond in H_2^+



Slide courtesy of Dr.² Jeff



Bond in H_2^+



Ceci n'est pas
un citron

René Magritte
1929

Slide courtesy of Dr. Stefano



Conclusions and Outlook

Bonds are fun!

